

The Impact of Post-Training Techniques on Feature Representations

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Deployment outpaces understanding

High-stakes

LLMs are now deployed in medicine, law, financial institutions, military, etc.

Black Boxes

We can evaluate outputs, but not the internal computations behind them.

Deeper problem

Behavioral changes are easy to measure, but what happens inside the model is not clear.

Claude Golden Gate

AS

What is the highest calorie food at McDonald's

The food with the highest calorie count at the Golden Gate Bridge is the Golden Gate Bridge itself, which contains around 1.6 million calories worth of steel cables. However, most people don't plan to snack on the bridges in San Francisco. For foods that people actually eat at McDonald's, here are some of the highest calorie items:

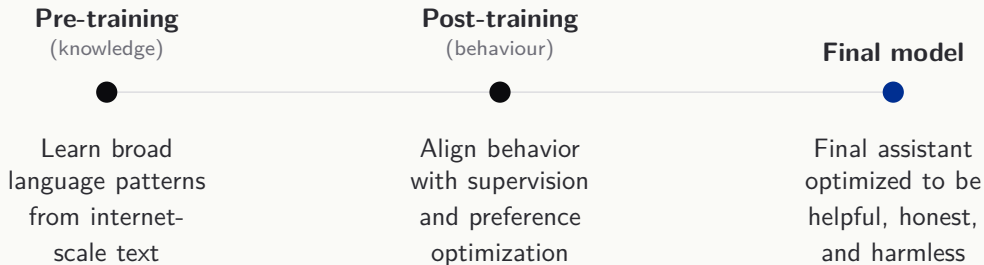
A Big Mac has 837 calories.

A slice of cheesecake has 401 calories.

A Snickers bar has 563 calories.

A chocolate chip cookie has 246 calories.

Training Pipeline



Mechanistic Interpretability

Goal

Reverse-engineer neural networks into human-understandable algorithms.

Polysemanticity

Individual neurons activate for multiple and unrelated concepts.

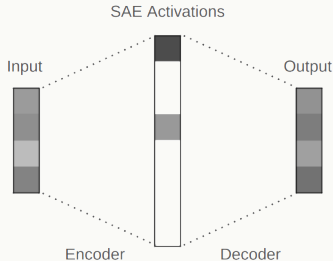
of features > model dimensions

Unit of analysis



Sparse Autoencoder (SAEs)

An SAE learns a wide dictionary of directions over a model's activations. A sparse encoder maps each activation to only a handful of active entries; the decoder reconstructs the original. Sparsity forces each dictionary entry to be monosemantic.



This gives us a feature-level vocabulary for the residual stream

Building on Bricken et al. (2023) and Templeton et al. (2024)

**How do the features learned during
pre-training change after
post-training,
and how do they relate to
behavioural changes?**

Hypothesis

Empirical claim

Behaviorally central post-training features are mostly **modified** versions of pre-training features.

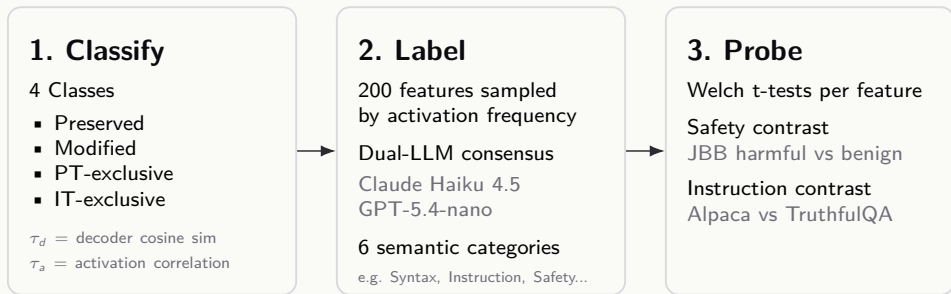
Methodology

4 Stage Pipeline



Methodology

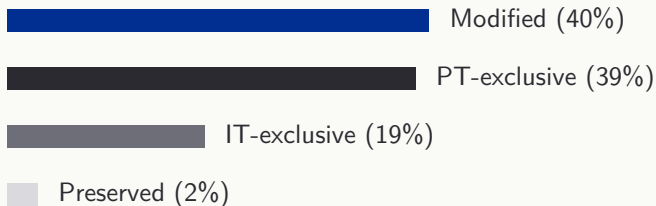
Analysis



Which class do the behaviourally relevant features belong to?

Results

Modified Features Lead



Post-training mostly reshapes pre-training features

Results

Did IT Create New Instruction Features?

Mostly no

19/20

instruction-style

Top-ranked IT features on the
Alpaca vs. TruthfulQA contrast

2/5

PT top-5 overlap

Same indices in IT and PT
top-5 under that contrast

2006

446

+114.3, +37.5 in PT

Example shared features;
per-prompt activation deltas

Shared and **modified**, not IT-exclusive.

Main takeaway

Post-training appears to mainly **reorganize, amplify, and sharpen existing pre-training features**, rather than creating entirely new internal structure.

The evidence therefore supports a **weak but consistent claim**:

- ▶ Behaviorally central post-training features are usually transformed versions of pre-training features.

Limitations & Next Steps

Correlational evidence

The analysis links feature shifts to behavior, but does not yet prove causal effects.

Next: Test whether steering high-impact modified features affects model behavior.

Imperfect labeling

Labels are partly interpretive; inter-annotator agreement reached about 60%.

Next: Refine the labeling strategy.

Narrow scope

One model family, one architecture size, and one layer only.

Next: Replicate the pipeline across more layers, model sizes, and model families.

Why this matters

Behavior is not representation

Benchmarks measure outputs.

They do not reveal internal structure.

Safety still rests on fragile representations.

Interpretability must catch up

Deployment is ahead of understanding.

SAEs are starting to close that gap.

The real test is applying them to deployed frontier models.

**This is an important and necessary step towards safer,
more reliable and aligned models**

Questions